Selecting Observations to Build Counterfactuals: A Cross-Disciplinary Literature Review

Methods in economics and the medical / epidemiological sciences, with electronic-health-record applications

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1. The problem and why "selecting observations" is the heart of it

Causal questions are counterfactual questions. Under the Neyman–Rubin potential-outcomes framework, each unit *i* has a pair of potential outcomes — $Y_i(1)$ under treatment and $Y_i(0)$ under control — but only one is ever observed (Rubin, 1974). This is the *fundamental problem of causal inference*: the counterfactual outcome is missing by construction. Every method reviewed here is, at bottom, a strategy for *imputing the missing counterfactual* by borrowing information from observed units that did not receive the treatment of interest (or received a different one). The central design decision is therefore **which observations to use, with what weight, to stand in for the counterfactual** — through discarding, matching, weighting, stratifying, or synthesizing comparison units.

Two largely separate research traditions converged on the same mathematics. In economics, the literature grew out of program evaluation and labor economics (the effect of job-training programs on earnings). In biostatistics and epidemiology, it grew out of observational comparative-effectiveness and drug-safety research. Imbens & Wooldridge (2009) and Stuart (2010) are the canonical bridges between the two vocabularies. The terminology differs — economists say "selection on observables," epidemiologists say "no unmeasured confounding," statisticians say "strong ignorability" — but the underlying assumption is identical.

Three assumptions recur throughout and govern whether *any* selection of comparison observations can recover a causal effect:

- **Unconfoundedness / ignorability / exchangeability**: conditional on measured covariates *X*, treatment assignment is independent of the potential outcomes (Rosenbaum & Rubin, 1983). All of the methods below operate *under* this assumption; none can manufacture it.
- **Overlap / positivity / common support**: every covariate profile that appears in the treated group must also have positive probability of appearing untreated, and vice versa. Without overlap there are simply no comparable observations to select.
- **SUTVA / consistency**: no interference between units and a single, well-defined version of treatment.

The choice of comparison observations also *defines the estimand* — the average treatment effect (ATE), the effect on the treated (ATT), the effect on the controls (ATC), or some overlap-weighted population. A recurring theme is that pursuing comparability often means quietly changing which population the estimate describes.

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2. The propensity score and the balancing-score theorem

The single most influential result is Rosenbaum & Rubin's (1983, *Biometrika*) demonstration that the **propensity score** — the conditional probability of treatment given covariates, $e(X) = \Pr(W = 1 \mid X)$ — is a *balancing score*. Conditioning on the scalar $e(X)$ is sufficient to remove all bias due to the observed covariates X, collapsing a high-dimensional matching problem into one dimension. This unlocked three families of adjustment that all amount to selecting/weighting comparison observations: matched sampling, subclassification (Rosenbaum & Rubin, 1984), and weighting (Rosenbaum, 1987). Subsequent work extended the score's role into doubly robust estimators such as augmented inverse-probability weighting (Robins & Rotnitzky, 1995) and targeted maximum-likelihood estimation (van der Laan & Rubin, 2006).

A crucial subtlety, developed largely on the economics side, is that the propensity score is usually *estimated*, and the first-step estimation materially changes the large-sample variance of any downstream matching estimator (Abadie & Imbens, 2016, *Econometrica*; Hirano, Imbens & Ridder, 2003).

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3. Matching: choosing discrete comparison units

Matching selects, for each treated unit, one or more control units judged "similar enough," discarding the rest. It is intuitive — it visibly produces comparable treated/control pairs (Rosenbaum & Rubin, 1983) — and it is non-parametric in the comparison it creates.

Exact and coarsened exact matching (CEM). Exact matching requires identical covariate values and is infeasible beyond a few discrete covariates. Iacus, King & Porro (2011/2012, *Political Analysis*) introduced **Coarsened Exact Matching**, the leading member of their *Monotonic Imbalance Bounding* class: continuous covariates are temporarily binned ("coarsened") as in drawing a histogram, exact matching is performed on the coarsened data, and the original values are then restored for analysis. CEM bounds the maximum imbalance *ex ante* by user choice, prunes units outside common support automatically, and (unlike many methods) guarantees that improving balance on one covariate does not worsen it on another.

Distance-based matching. Mahalanobis-distance matching and nearest-neighbor matching select controls minimizing covariate distance. Practical choices — caliper width, 1:1 vs. *k*:1 ratio, matching with vs. without replacement — trade bias against variance: matching with replacement and more neighbors lowers bias but raises variance (Abadie, Drukker, Herr & Imbens, 2004).

Propensity-score matching (PSM). Matching on $e(X)$ (Rosenbaum & Rubin, 1983, 1985) is the workhorse of applied medicine. Dehejia & Wahba (1999, *JASA*) famously showed that PSM applied to a comparable subset of the LaLonde (1986) job-training comparison group could approximately recover the experimental benchmark, while Smith & Todd (2005) argued the result was fragile to specification and sample — a debate that shaped the field's caution about matching.

Genetic matching. Diamond & Sekhon (2013, *REStat*) automate the choice of distance metric using a genetic search algorithm that iteratively reweights covariates to optimize balance, rather than relying on a single pre-specified propensity model.

Large-sample theory and bias correction (economics). Abadie & Imbens established the formal statistical properties matching had previously lacked: matching estimators with a fixed number of matches are *not* √N-consistent in general when more than one continuous covariate is used (2006, *Econometrica*); a bias correction restores √N-consistency and asymptotic normality (2011, *JBES*); the bootstrap fails for matching estimators (2008); and a martingale representation (2012, *JASA*) characterizes their variance.

The PSM critique. King & Nielsen (2019, *Political Analysis*) argue provocatively that PSM, by approximating a *completely* randomized experiment rather than a fully blocked one, can *increase* imbalance, model dependence, and bias (the "PSM paradox"), and recommend other matching methods (e.g., Mahalanobis or CEM) when matching is desired. The propensity score, they note, remains useful for weighting and subclassification. The point is contested (e.g., simulation comparisons find subclassification and weighting robust), but it crystallized a shift away from naive 1:1 PSM.

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4. Subclassification and stratification

Rather than matching pairs, subclassification partitions units into strata of similar propensity scores and compares treated and control within each stratum (Rosenbaum & Rubin, 1984). Roughly five strata remove the bulk of bias. **Fine stratification** uses many narrow strata and reweights, and performs competitively with weighting in low-prevalence, rare-outcome claims settings. Conceptually, stratification is weighting where every unit within a stratum receives the same weight.

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5. Weighting: keeping all observations, reweighting their contribution

Weighting reframes the selection problem: instead of discarding, every comparison observation is retained but contributes in proportion to how much it resembles the target population.

Inverse-probability weighting (IPW). Each unit is weighted by the inverse of its probability of receiving the treatment it actually got, reconstructing the covariate distribution of the target population (Horvitz–Thompson logic; Hirano, Imbens & Ridder, 2003, show that weighting by the *estimated* score is efficient). IPW's Achilles heel is extreme weights when propensity scores approach 0 or 1.

A unifying framework — balancing weights and overlap weights. Li, Morgan & Zaslavsky (2018, *JASA*) define a general class of *balancing weights* that re-weight each group toward an analyst-chosen target population, with IPW (ATE) and ATT weights as special cases. Their proposed **overlap weights** give each unit a weight proportional to its probability of being assigned to the *opposite* group [treated units get $1 - e(X)$, controls get $e(X)$]. Overlap weights are bounded (no extreme weights), minimize the asymptotic variance among balancing weights, and yield *exact* mean balance when the propensity score is fit by logistic regression. They target the population with the most covariate overlap — those for whom the data are most informative. Thomas, Li & Pencina (2020, *JAMA*) introduced overlap weighting to clinical audiences as a method that "mimics attributes of a randomized trial"; Li, Thomas & Li (2019, *AJE*) document its advantage under extreme propensity scores.

Balancing-by-design weighting. Rather than fit a model and hope it balances, these methods *solve directly* for weights that achieve balance. **Entropy balancing** (Hainmueller, 2012, *Political Analysis*) finds weights closest to uniform (minimum Kullback–Leibler divergence) subject to exactly matching specified covariate moments (means, variances, higher moments). **Stable balancing weights** (Zubizarreta, 2015, *JASA*) minimize weight variance subject to approximate balance constraints. The **covariate balancing propensity score** (Imai & Ratkovic, 2014, *JRSS-B*) estimates the score so that balance is built into the estimating equations. Zhao & Percival (2017) show entropy balancing is doubly robust. A caution from applied work: balancing weights can occasionally achieve balance only by placing extreme weight on a few observations (Springer HSORM evaluation, 2017).

Doubly robust estimation. Augmented IPW (Robins & Rotnitzky, 1995) and related estimators combine an outcome model with a propensity model and are consistent if *either* is correctly specified — a hedge against misspecifying the comparison.

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6. Restricting the comparison population: overlap, trimming, and common support

When overlap is poor, no clever weighting fully rescues the analysis, and the honest move is to change the population being studied. Crump, Hotz, Imbens & Mitnik (2009, *Biometrika*) formalize this: they characterize the *optimal subpopulation* for which the ATE can be estimated most precisely and show that, under broad conditions, the optimal rule is simply to discard units with propensity scores outside roughly [0.1, 0.9] ("moving the goalposts" by changing the estimand). Heckman, Ichimura & Todd (1997) had earlier shown that "comparing the incomparable" — violating common support — is a major source of evaluation bias. In the OHDSI medical ecosystem the analogous device is the **preference score**, a transformation of the propensity score used to assess empirical equipoise and trim non-overlapping patients before comparison (Schuemie et al.).

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7. Synthetic counterfactuals for aggregate units (the economics tradition)

When the treated unit is a single aggregate entity — a state, a country, a firm — there is no pool of similar treated units to average over, and matching individual observations is the wrong frame. Two influential approaches construct the counterfactual differently.

Synthetic control. Abadie & Gardeazabal (2003, *AER*) and Abadie, Diamond & Hainmueller (2010, *JASA*; 2015, *AJPS*) build a *synthetic* counterfactual as a data-driven **weighted average of "donor" units** chosen so that the weighted combination reproduces the treated unit's pre-intervention outcome trajectory and predictors. The canonical applications — the economic cost of terrorism in the Basque Country, California's Proposition 99 tobacco program, and West German reunification — illustrate the method's core contribution: a *transparent, data-driven rule for selecting and weighting comparison units*, replacing ad hoc choice of a single comparison region. Crucial design choices include curating the **donor pool** (e.g., excluding states that ran their own tobacco programs) and guarding against overfitting; inference relies on placebo (permutation) tests in space and time. Abadie's (2021, *JEL*) survey lays out the feasibility and data requirements.

Difference-in-differences and comparison-group selection. DiD compares the *change* in outcomes for a treated group against the *change* for a comparison group, identified under a *parallel-trends* assumption. The recent econometrics literature is precisely about *which comparison observations are admissible*. Goodman-Bacon (2021) and de Chaisemartin & D'Haultfœuille (2020, *AER*) showed that two-way fixed-effects regression with staggered adoption implicitly uses already-treated units as controls, producing biased "forbidden comparisons." Callaway & Sant'Anna (2021, *J. Econometrics*) re-found DiD on *group-time* average treatment effects, with comparison groups restricted to never-treated or not-yet-treated units, and allow parallel trends to hold only *conditional on covariates* via outcome-regression, IPW, or doubly-robust estimands. Sun & Abraham (2021) address the same problem in event studies. **Synthetic difference-in-differences** (Arkhangelsky, Athey, Hirshberg, Imbens & Wager, 2021, *AER*) unifies the two traditions, selecting *both* unit and time weights. Roth, Sant'Anna, Bilinski & Poe (2023) synthesize this fast-moving area.

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8. Design-stage selection in medicine: target trial emulation and comparator choice

The most important conceptual shift in medical causal inference is that *the selection of comparison observations begins before any statistical adjustment*, at the level of study design. Two complementary frameworks dominate.

Target trial emulation (TTE). Hernán & Robins (2016, *AJE*; Hernán, Wang & Leaf, 2022, *JAMA*) propose explicitly specifying the *hypothetical randomized trial* one would run — eligibility, treatment strategies, time zero, outcome, estimand — and then emulating each component in the observational data. TTE disciplines *which observations enter the comparison* (e.g., only eligible new initiators at a well-defined time zero) and prevents classic errors such as **immortal time bias** and prevalent-user bias (Hernán, Sauer, Hernández-Díaz, Platt & Shrier, 2016). Landmark demonstrations include emulating statin trials in UK primary-care EHR (Danaei et al., 2013) and reconciling the hormone-replacement-therapy discrepancy between observational studies and the Women's Health Initiative. TTE is now embedded in regulatory and HTA guidance (FDA Sentinel's framework; NICE; the Cochrane ROBINS tool) and has its own reporting standard, the **TARGET statement** (Cashin et al., 2025, *JAMA*).

New-user and active-comparator designs. Ray (2003, *AJE*) introduced the **new-user (incident-user) design**, restricting the sample to patients observed from treatment initiation so that covariates are measured before exposure and early events are not missed. Lund, Richardson & Stürmer (2015) and Sendor & Stürmer (2022) develop the **active-comparator, new-user (ACNU) design**: instead of comparing initiators to non-users, one compares initiators of the drug of interest to initiators of a clinically equivalent alternative for the *same indication*. By implicitly conditioning on indication, ACNU mitigates **confounding by indication** — the dominant threat in pharmacoepidemiology — and reduces healthy-user bias. The **prevalent new-user design** (Suissa et al.) extends the comparison to switchers. The trade-off is a smaller, sometimes less representative comparison pool. Selecting an appropriate active comparator is itself a substantive judgment that determines whether unmeasured confounders are balanced "for free."

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9. Selecting comparison observations in electronic health records

EHR and administrative-claims data are large, longitudinal, and messy: confounders such as disease severity, frailty, BMI, smoking, and lab values are often unmeasured or proxied imperfectly, and the data are generated by the very care process under study. Several methods were developed specifically to select and weight comparison observations in this setting.

High-dimensional propensity score (hdPS). Schneeweiss, Rassen, Glynn, Avorn, Mogun & Brookhart (2009, *Epidemiology*) proposed an algorithm that, rather than relying on a hand-picked covariate list, *empirically* generates thousands of candidate covariates from claims "dimensions" (diagnoses, procedures, drugs), assesses their recurrence, prioritizes them by their potential to confound (the Bross formula), and selects the top *k* (commonly ~500) into the propensity model. The premise is that many weak proxies collectively stand in for unmeasured confounders. In benchmark drug-safety studies (COX-2 inhibitors vs. NSAIDs and GI bleeding; statins and mortality) hdPS moved estimates toward randomized-trial expectations relative to adjustment for predefined covariates only. Rassen et al. (2023, *PDS*) provide ISPE-endorsed planning, implementation, and reporting guidance, and machine-learning/doubly-robust extensions now exist. A persistent open question is how to choose *k*: too few leaves residual confounding, too many overfits and inflates variance.

Large-scale propensity scores (LSPS). The Observational Health Data Sciences and Informatics (OHDSI) community, working on the OMOP Common Data Model, pushes this further: rather than selecting a few hundred covariates, LSPS fits an *L1-regularized (LASSO) logistic regression on tens of thousands* of baseline covariates spanning all prior conditions, drugs, and procedures (Tian, Schuemie & Suchard, 2018, *IJE*; implemented in the `CohortMethod`/`Cyclops` R packages). The LEGEND hypertension study (Schuemie et al., 2020, *JAMIA*) reported that LSPS matching achieved balance on *every one* of thousands of covariates for most comparisons under a stringent standardized-difference threshold — far more demanding than the handful of covariates typically balanced in conventional studies — across a multi-country network of EHR and claims databases.

Negative controls and empirical calibration. Because residual confounding is expected even after careful comparison-group selection, OHDSI pairs estimation with diagnostics that *use the data to detect remaining bias*. **Negative controls** are exposure–outcome pairs with no plausible causal relationship; running the same analysis on dozens of them reveals the systematic-error distribution that the design fails to remove. **Empirical calibration** then adjusts p-values and confidence intervals to reflect both random and systematic error (Schuemie, Ryan, DuMouchel, Suchard & Madigan, 2014; Schuemie et al., 2016, *Stat. Med.*; 2018, *PNAS*). In LEGEND, calibration brought 95% CI coverage close to the nominal level, versus the wide miscoverage seen in much published observational work. Selecting good negative controls is itself an active sub-problem (Voss et al.; the Common Evidence Model), and *synthetic* negative controls have been proposed to screen analyses before they are run (Fortin, Johnston & Schuemie, 2021).

Worked EHR exemplars. Beyond LEGEND and the UK statin emulations, recent EHR-based counterfactual studies include COVID-19 vaccine comparisons via TTE in the US Veterans Affairs system (Ioannou et al., 2022), the Israeli third-dose effectiveness study (Barda et al., 2021), federated TTE across hospital systems (eICU/MIMIC-IV and the INSIGHT NYC network, 2025), and high-throughput TTE for Alzheimer's drug repurposing using the RECOVER and INSIGHT data. A practical lesson repeatedly emphasized is the tension between **EHR continuity / completeness** and sample size: restricting to patients with denser records reduces misclassification and bias but shrinks the comparison pool.

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10. Machine-learning and doubly-robust estimators

A newer line lets flexible learners model the treatment and/or outcome surfaces while retaining valid inference for the causal effect, increasingly applied to high-dimensional EHR covariates.

- **Double/debiased machine learning (DML)** (Chernozhukov, Chetverikov, Demirer, Duflo, Hansen, Newey & Robins, 2018, *Econometrics Journal*): uses ML to flexibly adjust for many confounders, then removes regularization bias via Neyman-orthogonal scores and cross-fitting, achieving $\sqrt{N}$-consistent ATE estimates. Unlike matching, it need not discard data and handles continuous covariates naturally.
- **Targeted maximum-likelihood estimation (TMLE)** (van der Laan & Rubin, 2006; van der Laan & Rose, 2011): a semiparametric, doubly-robust plug-in framework that "targets" an initial outcome fit using propensity information.
- **Causal forests / generalized random forests** (Wager & Athey, 2018, *JASA*; Athey, Tibshirani & Wager, 2019, *Ann. Statist.*): adapt random forests to estimate *heterogeneous* (conditional) treatment effects with honest splitting and valid confidence intervals; the comparison is constructed locally, among covariate-similar units.

- **Bayesian Additive Regression Trees (BART)** (Hill, 2011, *JCGS*): flexible outcome modeling for causal effects.
- **Meta-learners (S-, T-, X-learners)** (Künzel, Sekhon, Bickel & Yu, 2019, *PNAS*): wrap any supervised learner to estimate the CATE.

These methods relax functional-form assumptions about *how* one adjusts for confounders but cannot relax unconfoundedness itself, and naive plug-in of ML predictions without de-biasing yields invalid inference.

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11. Cross-disciplinary synthesis

The economics and medical literatures emphasize different things while sharing the same potential-outcomes core (Imbens & Wooldridge, 2009; Stuart, 2010; Imbens & Rubin, 2015).

- **Unit of analysis.** Economics often works with *aggregate* treated units (one state, one country), motivating synthetic control and synthetic DiD. Medicine works with *many micro units* (patients), motivating matching, weighting, and high-dimensional propensity scores.
- **Design vs. analysis emphasis.** Pharmacoepidemiology front-loads selection into the *design* (target trial, new-user, active-comparator), treating the comparison group as a design object. Econometrics historically emphasized estimator properties and identification, though the target-trial logic is now diffusing into applied economics, and the DiD revolution is essentially a debate about admissible comparison observations.
- **Diagnostics.** Medicine, especially OHDSI, has institutionalized empirical-bias diagnostics (negative controls, calibration) at a scale rarely seen in economics; economics has more developed formal sensitivity-to-design and inference theory (placebo tests, randomization inference).
- **Shared cautions.** Both fields converged on covariate balance as the design target, common support / overlap as a precondition, and the recognition that the choice of comparison observations silently fixes the estimand.

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12. Open challenges

1. **Unmeasured confounding.** No selection method overcomes confounders absent from the data. Sensitivity analysis is essential: Rosenbaum's bounds, and the E-value (VanderWeele & Ding, 2017, *Ann. Intern. Med.*) for medical studies, quantify how strong an unmeasured confounder would need to be to overturn a result.
2. **Positivity in high dimensions.** As covariate sets grow (hdPS, LSPS), strict overlap becomes harder to satisfy, and extreme weights or empty matching strata proliferate. Overlap

weights and principled trimming are partial answers.

3. **EHR data quality.** Informative observation/visit processes, measurement error and misclassification in codes, missing eligibility data, and EHR discontinuity all bias comparison-group construction in ways adjustment cannot fully repair.
4. **Time-varying treatment and confounding.** Sustained-treatment strategies require g-methods (marginal structural models with IPW; g-computation) layered on top of comparison-observation selection.
5. **Generalizability / transportability.** A counterfactual built from one population's comparison units may not transport to another; reconciling internal validity (comparability) with external validity (representativeness) is unresolved, and is part of why estimands like the overlap-weighted ATO are scrutinized.
6. **Estimand transparency.** Because each selection rule implies a different target population, reporting standards (TARGET, 2025) increasingly demand that authors state the estimand and the population their selected observations actually represent.

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Note on scope: This review concentrates on methods that select, weight, or synthesize **comparison observations** under unconfoundedness/parallel-trends-type assumptions. It deliberately gives lighter treatment to designs that exploit exogenous variation rather than observed-covariate comparability — instrumental variables, regression discontinuity, and selfcontrolled designs (case-crossover, SCCS) — which select counterfactual *information* through identification strategies rather than through choosing comparable units. These are natural extensions if you want the review broadened.